

Asad Ikram

Lahore, Pakistan · asad.ikram53@gmail.com · +44 7482 376417 · linkedin.com/in/asad-ikram98 · github.com/asadfix

Dr Johannes Wachs

Center for Collective Learning, Corvinus Institute of Advanced Study

Corvinus University of Budapest, Hungary

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Re: Research Assistant / PhD Student positions, Geography of Code group

Dear Dr Wachs,

I have written software remotely for companies in six countries, and in 2026 most of the code I ship is written by AI agents I direct. It started in December 2018 with small data jobs on Fiverr while I was studying computer science in Pakistan, more than 300 completed orders, and grew into contract work for companies in Canada, Italy, Ireland, the United Arab Emirates, the United States and the United Kingdom. This year the work itself changed. At M+C Saatchi Fluency I keep a fleet of forum crawlers running for a global automotive brand. When a forum changes its pages, an LLM agent I designed diagnoses the break, rewrites the crawler and tests it on live pages, and I decide whether the change merges. Most of the rest of that code, including the pipeline that builds new crawlers, is written by coding agents to my specification and checked by me. My job has moved from writing code to deciding what code should exist and whether it is right. Your group studies how platforms and AI change where and how skilled software work is done, and I would like to spend the next years measuring that change rather than only living it.

Your work. The paper I keep returning to is *Who is using AI to code?* with Daniotti, Feng and Neffke (*Science*, 2026). A classifier for AI-written Python, applied to more than 30 million GitHub commits, estimates that AI wrote about 29% of US Python functions by the end of 2024 and raised output by about 3.6%, with the gains going to experienced developers while early-career developers, the keenest users, showed no significant benefit. That fits what I see: an agent saves me time only because years of writing such code by hand let me judge its output quickly. The paper is open about its limits, open-source Python on GitHub in six countries, and asks for countries at different income levels. Your map of Claude Code on GitHub shows a related limit. It counts commits carrying Claude Code's default co-author trailer. My public commits carry it, but my profile gives no location and most of the agent-written code I ship goes into private client repositories, so a developer like me is largely invisible to it. With del Rio-Chanona and Laurensyeva (*PNAS Nexus*, 2024) you showed that Stack Overflow activity fell by about 25% within six months of ChatGPT, relative to sites where it was less available or less capable: the public traces we study are themselves changed by the technology under study. And *The Geography of Open Source Software* (2022) and *Digital Traces of Brain Drain* (2023) show how far GitHub locations can go in tracking where developers are and where they move.

A question I could contribute to. How do AI coding agents change the division of labour in software work, and for whom? Agents now leave signatures in public traces: co-author trailers, bot accounts that open pull requests, and the code patterns your classifier detects. Following developers' commits and collaboration networks before and after they start working with agents, one could ask whether agent use changes who leads a project and the uneven workloads that your paper with Betti, Gallo and Battiston (*Nature Communications*, 2025) links to success, whether review work shifts towards senior developers, and whether developers in lower-income countries, including the many who, like me, enter global software production through freelance platforms and remote contracts, see the gains your *Science* paper finds. I would bring one habit from my own research: asking how much of a measured change comes from the platform's visibility rules rather than from behaviour. A trailer that is on by default for one tool and absent for another, a blank profile location, or work moving into private repositories can each look like a change in adoption. I would build that into the design, for example by checking trace-based measures against repositories where the true agent use is known, starting with my own.

Coding, modelling and data wrangling. Python and SQL are my working languages, with PySpark and AWS when data are large and PostgreSQL, Redshift and Athena as the databases I use most. At Dubizzle Labs in Lahore (February 2022 to August 2024, promoted twice to Senior Data Engineer) I led three engineers running more than 500 scrapers across 15 markets in English, Arabic, Urdu and Hindi, redesigned the layered warehouse,

and built competitor dashboards from listings and online discussion that predicted revenue. At Fluency, where I have been a part-time Data Engineer since December 2024, I designed the fleet of 64 crawlers and operate a weekly pipeline for a global e-commerce client whose master table holds 2.2 million rows by 184 columns, with an 89-check quality sweep before each release. At Fix.com, a Canadian retailer I have contracted for since May 2023, I designed and built its crawler platform on my own this year, about 10 million rows across 82 datasets. Joining, deduplicating and checking messy data at scale before anyone models it is the part of empirical work I am most practised in.

Research so far. I hold an MSc in Data Analytics with Distinction from the University of Portsmouth, where I was a Chevening Scholar in 2024/25. My dissertation analysed public reaction to UK economic policies in about 279,000 posts from four platforms, with a fine-tuned BERT classifier against baselines, McNemar and bootstrap tests and Prophet forecasting. Its weakness taught me most: the model was validated against labels seeded by a lexicon, so the check was partly circular. That led to a manuscript with my supervisor, Dr Ella Haig, *Is the Platform Part of the Measurement?*, complete and not yet published, which treats each platform as a survey mode. In a 500-replication simulation I built, pooling platform scores at face value gives the wrong sign of daily aggregate sentiment on roughly a quarter of days. The code is public (github.com/asadfix/platform-measurement, DOI 10.5281/zenodo.23105129). At Artemis AI, a pre-launch analytics start-up where I am co-founder and CTO in a team of eight, I found that our reported classifier accuracies measured agreement with the LLM that produced the training labels rather than with people, and we are rebuilding evaluation around a human-labelled set. What a number actually measures is the question I would bring to GitHub data.

What I do not yet bring. I have no peer-reviewed publications. My statistics are applied: I have used hypothesis tests, the bootstrap and forecasting, but I have not been trained in panel econometrics or in the difference-in-differences and event-study designs your papers use, and I have used network tools in practice without formal training in network science. These are what I would want the doctoral programme in Economics or Business Informatics, and your group, to teach me, and I would start on them as a research assistant well before Fall 2027.

Practicalities. I could start on 1 December 2026 or later, and I would relocate to Budapest. I currently hold part-time roles, at Fluency, at Fix.com and as co-founder of Artemis AI, and I would like to discuss openly with you how the full-time position fits with them. My referees are Dr Ella Haig, my MSc supervisor and co-author, and Dr Fahad Ahmad, who taught and assessed me on the MSc, both at the University of Portsmouth; their contact details are in my email and CV. Thank you for considering my application.

Yours sincerely,

Asad Ikram